

Process or Transmit? An Energy- and Thermal-Aware Onboard Decision Engine for LEO Satellite Data Handling

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Abstract—Small Earth-observation satellites capture multi-spectral and Synthetic Aperture Radar data under intermittent contact and constrained onboard resources. This paper studies the IASTAM 6.0 “Process or Transmit?” problem using a seeded, one-second-step LEO simulator and its implemented threshold heuristic. Across five three-orbit episodes (seeds 42–46), the heuristic achieved a mean delivered-utility ratio of 0.0289 (sample standard deviation 0.0039; 95% t interval 0.0241–0.0337) and completed 1.57% of generated payloads. Mean simulated energy draw was 137.36 kJ, mean battery state of charge stayed above the 30% floor, and no safe-mode entry occurred. In a separate two-orbit trace (seed 42), simulated chip temperature reached 52.48°C; the simulator throttles compute at 50°C but does not enforce a hard temperature cap. Learned-policy, Lyapunov-scheduler, oracle, and hardware-calibration studies remain future work.

Index Terms—orbital edge computing, onboard task scheduling, energy- and thermal-aware computing, Lyapunov optimization, reinforcement learning, Low Earth Orbit, fault tolerance.

I. INTRODUCTION

The 2026 edition of the IEEE IAS Tunisia Annual Meeting (IASTAM 6.0) Technical Challenge, organized in collaboration with the Tunisian Space Association (TUNSA), addresses an operational bottleneck in Earth-observation data handling. Orbital edge computing has drawn attention as a way to move selected processing closer to sensors [1], [8]. Orbital testbeds such as Starcloud’s spaceborne GPU demonstrator [10] and Google’s Project Suncatcher concept [8] underscore that orbital edge computing is now an operational engineering reality.

Within this domain, Problem 1 (“Process or Transmit?”) targets an acute bottleneck in Earth Observation (EO): small satellites continuously capture high-resolution optical and Synthetic Aperture Radar (SAR) payloads (generating multi-megabyte payloads every 15 s), yet operate under three binding physical constraints:

- 1) **Battery Floor:** The simulator models a 100 Wh battery and clamps its state at a 30% minimum depth-of-discharge floor.

- 2) **Thermal Constraint:** Vacuum requires radiative heat rejection. The current simulator uses a first-order thermal proxy and pauses compute at 50°C; it does not enforce a hard cap, and the measured trace reaches 52.48°C.
- 3) **Severe Downlink Bandwidth Choke:** Line-of-sight contact with ground stations is limited to short windows (120–180 s per 90-minute orbit, *about 2.8% mean duty* contact duty cycle) with an X-band rate of 20.0 Mbps (2.5 MB/s).

For every captured payload, the onboard system must choose among four actions: PROCESS_NOW (compression vs. inference), STORE_QUEUE (time-shifted execution in Mass Memory Unit, MMU), TRANSMIT_RAW (downlink during line-of-sight passes), or DISCARD_DROP (pruning low-utility or cloud-occluded data). The optimization objective is to maximize cumulative scientific and operational utility, which decays exponentially over latency.

Contributions:

- We characterize the repository’s seeded one-second simulator and implemented multi-threshold heuristic.
- We report five three-orbit heuristic episodes with decision quality, task completion, energy, latency, MMU/RAM utilization, SEU counts, and safe-mode events.
- We identify an observed temperature excursion above the code’s 50°C throttling threshold, a limitation that motivates thermal-model and safety-guard validation.

II. RELATED WORK

A. Orbital and Edge Computing Systems

Surveys of orbital edge computing delineate the compelling systems-level case for in-space processing: edge inference on commercial off-the-shelf (COTS) and radiation-tolerant hardware substantially mitigates downlink choke [1]. Serverless and cloud-native architectures for multi-tenant orbital platforms, such as Trabant [2] and studies on the Tiansuan constellation [3], demonstrate that containerized workloads can be time-shifted across orbital resource cycles. Broader treatments of Space AI [4] underscore the necessity of autonomous onboard classification, while Earth observation surveys [5] catalog compression and neural inference tradeoffs. Furthermore, space-grade FPGA neural accelerator benchmarks [6] and

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analytical studies on when to compute in space [7] establish the empirical energy-latency operating points adopted in our modeling.

B. Control and Scheduling Paradigms

Resource-constrained orbital scheduling is addressed via two dominant paradigms:

- 1) **Lyapunov Optimization & Drift-Plus-Penalty:** Prior work [11] motivates online scheduling; implementing and evaluating an orbital scheduler remains planned work.
- 2) **Constrained Reinforcement Learning (CMDP):** The repository includes a Gymnasium environment; policy training and safety evaluation are not reported here.

C. Added Value of This Work

Prior work motivates orbital edge processing and constrained scheduling. This interim study contributes a reproducible characterization of the current heuristic and identifies model and benchmarking gaps.

III. METHODOLOGY

A. Physical Models & Orbital Mechanics

The satellite orbital environment and hardware state at time t are formalized in Table I.

TABLE I
PRIMARY SYSTEM AND STATE VARIABLES

Symbol	Physical Definition	Nominal Value / Range
$\tau_{\text{orbit}}, \tau_{\text{sun}}$	Orbit period / Sunlit duration	5400 s / 3300 s (90 / 55 min)
$F_{\text{harvest}}(t)$	Solar power harvested	0.0 W (ecl) to 30.0 W peak
$\text{SoC}(t)$	Battery State-of-Charge	Floor ≥ 0.30 , Cap: 100.0 Wh
$T_{\text{chip}}(t)$	Processor temperature	Ambient 0°C , Ceiling $\leq 50.0^\circ\text{C}$
$G(t)$	Ground station contact flag	$\{0, 1\}$, 120–180 s pass (<i>about 2.8% meanduty</i>)
B_{downlink}	RF communication speed	20.0 Mbps = 2.5 MB/s (15.0 W)
P_{seu}	SEU bit-flip rate	10^{-5} s^{-1} cosmic radiation
$M_{\text{mmu}}, M_{\text{ram}}$	Mass Memory / RAM	32 000 MB / 4 000 MB

1) **Power and Battery Dynamics:** The simulator uses a 55-minute sunlight interval followed by a 35-minute eclipse. In sunlight, harvested power follows a sine-shaped profile from a 22.5 W base to a 30 W peak, with 30 s cosine tapers at sunlight boundaries; eclipse harvesting is zero.

Battery energy is updated by

$$E_{\text{batt}}(t+1) = \min(E_{\text{cap}}, E_{\text{batt}}(t) + [P_{\text{harvest}}(t) - P_{\text{draw}}(t)]\Delta t), \quad (1)$$

where $P_{\text{draw}} = P_{\text{base}} + P_{\text{compute}} + P_{\text{comm}}$, $P_{\text{base}} = 5 \text{ W}$, and radio draw is 15 W during downlink. The implementation has no charge/discharge efficiency; it clamps battery energy to the 30% floor on brownout and exits safe mode above 1.05 times that floor.

2) **Lumped Thermal Proxy:** The simulator uses a first-order thermal proxy rather than a Stefan–Boltzmann radiation model:

$$T_{\text{chip}}(t+1) = T_{\text{chip}}(t) + 0.15P_{\text{draw}}(t) - 0.10(T_{\text{chip}}(t) - T_{\text{amb}}). \quad (2)$$

This one-second update uses watts of total bus, compute, and communication draw, with $T_{\text{amb}} = 0^\circ\text{C}$. Compute pauses at or

above 50°C , but this threshold is not a hard cap; the seed-42 two-orbit trajectory peaks at 52.48°C .

B. Workload, Queueing, and Utility Formulation

Payloads k arrive periodically every $\Delta t_{\text{gen}} = 15 \text{ s}$ across dual modalities:

- **Optical Multispectral (70%):** Size $S_k \in [5, 15] \text{ MB}$, initial utility $U_{0,k} \in [10, 100]$.
- **Synthetic Aperture Radar (30%):** Size $S_k \in [100, 300] \text{ MB}$, initial utility $U_{0,k} \in [10, 100]$.

Payload utility decays exponentially over latency ($t - t_{\text{capture}}$):

$$U_k(t) = U_{0,k} \cdot \exp(-\alpha_k(t - t_{\text{capture}})) \quad (3)$$

with decay parameter $\alpha_k \in [10^{-4}, 10^{-3}] \text{ s}^{-1}$.

Compute actions transform payload properties:

- **Compression:** Duration 2 s, $P_{\text{comp}} = 10 \text{ W}$, size factor $C_{\text{ratio}} = 0.30$, utility retention $\rho_{\text{val}} = 0.90$.
- **Inference:** Duration 5 s (with 2 s wake penalty), $P_{\text{inf}} = 20 \text{ W}$ (30 W wake), size factor $C_{\text{ratio}} = 0.001$, utility retention $\rho_{\text{val}} = 0.50$.

C. Decision Engine & Solution Schedulers

For each payload k , the agent selects an action $a_k \in \mathcal{A}$, where:

$$\mathcal{A} = \{\text{PROCESS_COMPRESS, PROCESS_INFER, DOWNLINK_RAW, STORE_QUEUE, DISCARD_DROP}\} \quad (4)$$

- 1) **Baseline Heuristic Policy:** When $G(t) = 1$, downlinks at most three payloads, ranking processed payloads ahead of raw payloads and then by current utility. When $\text{SoC} > 0.50$, $T_{\text{chip}} < 50^\circ\text{C}$, and queue depth < 2 , it executes inference (for payloads $< 50 \text{ MB}$) or compression. Stale payloads ($U_k(t) < 5.0$) are evicted when MMU usage $> 80\%$.
- 2) **Constrained MDP (Gymnasium):** State $s_t \in \mathbb{R}^{13+5K}$ captures normalized telemetry and top- K payload metadata. Invalid action masks zero out physically infeasible actions.
- 3) **Planned Lyapunov Scheduler:** Drift-plus-penalty scheduling is a Phase 3 target and is not implemented in the current repository.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup and Results

The implemented heuristic was evaluated for five three-orbit episodes (16,200 one-second steps per episode; seeds 42–46), generating 1,080 payloads in each episode. Table II reports means, sample standard deviations, and 95% Student- t confidence intervals ($n = 5$). The evaluator’s decision-quality denominator is the snapshot sum of generated payload base values. Memory utilization is time-averaged occupancy and energy is cumulative modeled power draw.

TABLE II
IMPLEMENTED HEURISTIC ACROSS FIVE THREE-ORBIT EPISODES

Metric	Mean	Sample SD	95% CI
Decision quality	0.0289	0.0039	[0.0241, 0.0337]
Completed-task rate	0.0157	0.0020	[0.0133, 0.0182]
Cumulative energy (kJ)	137.364	1.675	[135.285, 139.443]
Mean latency (s)	7951.51	15.81	[7931.88, 7971.13]
MMU utilization (%)	28.124	1.799	[25.890, 30.357]
RAM utilization (%)	0.172	0.003	[0.169, 0.175]
SEU events / episode	0.200	0.447	[-0.355, 0.755]
Safe-mode events / episode	0.000	0.000	[0.000, 0.000]

The SEU confidence interval’s negative lower endpoint is a normal-approximation artifact for a nonnegative count. Across a separate two-orbit seed-42 trace, SoC ranged from 92.37% to 100%, while temperature ranged from 0.75°C to 52.48°C. The 50°C threshold throttles processing but does not bound the simulated temperature. Five short episodes also do not establish a statistically meaningful SEU recovery rate.

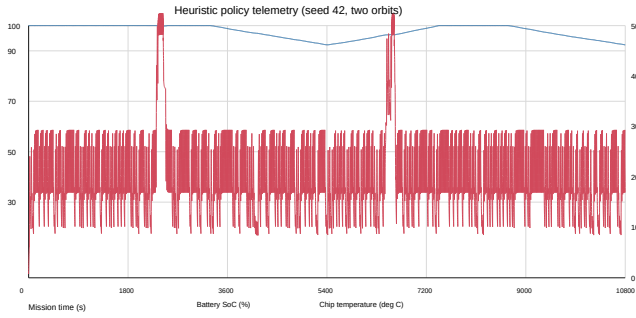


Fig. 1. Heuristic trajectory over two orbits (seed 42). Temperature exceeds the 50°C throttle threshold.

This is a single-policy characterization. The current code-base does not implement the claimed transmit-all comparison, Lyapunov scheduler, or MILP oracle; comparative policy results are future work.

B. Phase 3 Validation Plan

Phase 3 development will: (1) train a MaskablePPO agent in *simulation/rl_env.py*; (2) integrate the Lyapunov safety filter to bound RL proposals; (3) compute the MILP oracle ceiling; and (4) stress-test under solar flare SEU bursts ($p_{\text{seu}} = 10^{-3}$) and degraded battery packs (60 Wh).

V. CONCLUSION

We report a reproducible five-seed characterization of the implemented heuristic. The measured decision-quality ratio is 0.0289 on the current simulator workload. The two-orbit trace also exposes a thermal-model limitation: simulated temperature can exceed the 50°C throttling threshold. Future work includes comparative baselines, a calibrated thermal model with a hard safety guard, and implementation of Lyapunov and oracle schedulers.

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